

Following the Entitlement to the Permit: Linking Planning Commission decisions to DBI's building-permit record

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Abstract

A Commission decision is only half of what we want to know. The other half is what happened to the project: whether the permit issued, whether anything was built, how long it took, whether the applicant gave up. This memo establishes how far the item-level minutes dataset can be followed into the city's building-permit record, and what that buys.

Three results. First, **the permit-number join works at 97.6%** — not on a sample, on all 2,295 distinct permit numbers the minutes print, against all 1,295,048 rows of DataSF's Building Permits table. It is best for the oldest numbers, not worst. Second, **the apparent collapse of the permit route after 2019 was a parsing failure, not a data failure**. Around 2017 the minutes changed how they print a permit number, from 2013.10.21.9832 to 2019.0618.3775; discretionary review kept citing a permit on 95–100% of items throughout. Reading the new form recovers 711 items and 483 permits, and restores the join for 2017–2026. Third, **the permit route reaches one part of the docket and not the rest**: 85.5% of discretionary reviews carry a parsable permit number against 0.3% of conditional uses. For the rest, a validated block-and-lot fallback narrows a median-ten parcel history to a median-one candidate set at 0.72 precision and 0.92 recall.

One negative result, and it matters for what comes next: the Commission's *conditions of approval* are not in DBI. Across 1.3 million permit records, 87 — 0.007% — mention a condition of approval or a Commission motion at all. Whatever recovers condition text, it is not this dataset.

Contents

1	What is being joined	2
2	The permit numbers parse, and they match	2
3	The 2019 cliff was ours, not the archive's	4
4	Which parts of the docket the permit route reaches	6
5	What the match buys	6
6	The parcel route, scored honestly	7
7	Conditions of approval are not in DBI	8
8	What this establishes, and what it does not	9
9	Reproducing this	9

1 What is being joined

The item table is the extraction described in the corpus memo: 16,199 items heard by the San Francisco Planning Commission between 1998 and mid-2026 that carry a case number, in 29 structured fields, at 96.6% field-level accuracy on a frozen test half. Nothing here re-estimates that; this memo asks only what the table can be joined to.

The other side is DataSF’s **Building Permits** dataset (i98e-djp9), the Department of Building Inspection’s permit record: **1,295,048 rows** covering filings from 1901 to the present, with the permit’s parcel, type, description, status, cost, unit counts and five dates. It is downloaded once and cached, because a match rate reported on a sample is a match rate with a standard error, and the whole point of this exercise is to stop sampling.

Two facts about that table shape everything that follows.

permit_number is not a key. The 1,295,048 rows carry 1,149,247 distinct permit numbers: 145,795 rows are a repeat of a number already in the table, usually one row per lot the permit touches. The table is therefore collapsed to the earliest-filed row per number before joining. A join that ignores this double-counts multi-lot projects, which are exactly the large projects one cares about.

DBI does not zero-pad the serial. The permit the minutes print as 2000/01/19/431 is 20000119431 — eleven digits, not twelve. Stripping non-digits and comparing is therefore the right normaliser, and padding to a fixed width is the wrong one. Letter prefixes (M658747) and suffixes (9801703S) are stripped for the same reason.

2 The permit numbers parse, and they match

4,083 items mention a building, site or demolition permit in the request text. A number is recovered from **3,588** of them — 22.1% of the whole docket — giving **2,295 distinct permit numbers** in four printed forms.

Table 1: Permit numbers as the minutes print them, and whether DataSF’s Building Permits table (i98e-djp9, 1,295,048 rows) holds them. Distinct numbers, not items: a number cited at several hearings is counted once.

Printed form	Example	Distinct	Matched	Rate
YYYY.MM.DD.NNNN	2000/09/25/1401	1,529	1,488	97.3%
YYYY.MMDD.NNNN	2001-0227-3042	490	480	98.0%
packed	200006213244	42	41	97.6%
seven-digit	No. 9806249	234	230	98.3%
All		2,295	2,239	97.6%

Table 1 reports the match against DBI. **2,239 of 2,295 numbers — 97.6% — are present**, and this is a census of the parsed numbers rather than a sample of them. Table 2 breaks it out by decade of the hearing. The prior worth correcting here is that the pre-2001 seven-digit form would be the least certain, because it is the oldest and DBI’s coverage of that era is untested. It is in fact the *best* covered — 98.3% of the seven-digit numbers, and 98.7% of every number cited at a 1990s hearing. DBI’s table holds filings in bulk from the early 1980s, and the 1998–2000 minutes print the application number cleanly after “No.”. No decade falls below 97.1%.

Table 2: Match rate by decade of the hearing at which the number was cited. The pre-2001 seven-digit form is the *best* covered, not the worst.

Decade	Distinct permits	Matched	Rate
1990s	158	156	98.7%
2000s	1,242	1,209	97.3%
2010s	619	606	97.9%
2020s	276	268	97.1%

Table 3: Which parts of the docket the permit route reaches. ‘Cites a permit’ is any mention of a building, site or demolition permit in the request text; ‘number parsed’ is a number recovered from it. Request types with fewer than 40 items are omitted.

Request type	Items	Cites a permit	Number parsed	Matched in DBI
<code>discretionary_review</code>	4,117	95.2%	85.5%	83.8%
<code>conditional_use_modification</code>	295	3.1%	2.0%	2.0%
<code>downtown_project</code>	321	4.0%	1.9%	1.9%
<code>variance</code>	1,172	1.7%	1.6%	1.5%
<code>informational</code>	587	1.4%	0.7%	0.7%
<code>other</code>	459	3.1%	0.7%	0.4%
<code>conditional_use</code>	5,074	0.4%	0.3%	0.3%
<code>appeal_preliminary_negative_declaration</code>	537	0.7%	0.0%	0.0%
<code>ceqa_environmental</code>	652	0.3%	0.0%	0.0%
<code>general_plan_amendment</code>	334	0.0%	0.0%	0.0%
<code>historic</code>	177	0.0%	0.0%	0.0%
<code>large_project_authorization</code>	308	0.6%	0.0%	0.0%
<code>office_allocation</code>	301	0.3%	0.0%	0.0%
<code>planning_code_amendment</code>	1,301	4.0%	0.0%	0.0%
<code>rezoning_map_amendment</code>	261	1.1%	0.0%	0.0%

Table 4: What the match buys: what is observable in DBI for a matched item, on the far side of the hearing.

Observable	DBI field	Coverage
Permit ever issued	<code>issued_date</code>	78.5%
median lag, hearing → issuance	<code>issued_date</code>	245 days
90th percentile lag	<code>issued_date</code>	768 days
Construction started	<code>first_construction_document_date</code>	38.5%
Permit completed	<code>status</code>	48.9%
Withdrawn, expired, cancelled or revoked	<code>status</code>	32.9%
Cost recorded	<code>revised_cost / estimated_cost</code>	88.8%
median revised cost	<code>revised_cost</code>	\$350,000
Proposed dwelling units	<code>proposed_units</code>	84.8%
agrees with a unit count in the request text	<code>proposed_units</code>	57.1% of 42
Supervisor district	<code>supervisor_district</code>	99.9%
Matched item–permit pairs	---	3,801

Table 5: The parcel route, scored against the permit-number subset. ‘Coverage’ is the share of those items for which the rule returns any candidate; precision is computed on the items where it does; recall is the share of the truly cited permits it retains. $n = 3274$ items where the cited permit is present at the parcel DBI records.

Rule (block + lot, then ...)	Coverage	Precision	Recall	Median set
every permit at the parcel	100.0%	0.14	1.00	10
major types only, no date window	99.7%	0.44	0.98	3
major types, filed within $[-1, +0.5]$ yr	55.9%	0.75	0.52	1
major types, filed within $[-2, +1]$ yr	86.2%	0.72	0.83	1
major types, filed within $[-3, +1]$ yr	94.0%	0.72	0.92	1
... nearest filing only	94.0%	0.74	0.67	1
... nearest three filings	94.0%	0.72	0.89	1

Table 6: Reach of the parcel route on the part of the docket the permit-number route cannot see.

Items	Count	Share of the 16,199
No permit number in the text	12,611	77.9%
of which: block and lot present	10,076	62.2%
parcel exists in DBI	9,455	58.4%
a candidate permit in window and type	6,664	41.1%

All 56 misses were inspected rather than assumed. Every one is a plausible permit citation sitting immediately after a permit phrase in the source text — a transcription digit wrong in the minutes (one is dated “2019.05.90”), or an application that never reached the DBI table. None is a parser false positive.

Getting to that required one guard, and it is the reason the seven-digit form is reported separately. That pattern is only “No. followed by seven to nine digits”, and the minutes print several other numbers that way: notices of violation, DBI complaint numbers, a State Clearinghouse number for an EIR. Eight of them were entering the join as permits. The match is now kept only where the seventy characters before it name a permit and do not name one of the impostors, which removes seven numbers and raises the form’s match rate from 95.9% to 98.3% — the removed numbers were not permits, so they were never going to match.

3 The 2019 cliff was ours, not the archive’s

Before this memo, the parsed-permit counts thinned to nothing after 2019: on discretionary reviews, 9 items in 2020, 1 in 2021, 3 in 2022. Three explanations were possible — the modern minutes dropping permit numbers from the text, the collapse of discretionary review removing the items that carry them, or a parsing failure on a newer printed format. Figure 1 separates them, and it is the third.

The blue line is the share of discretionary-review items that mention a building permit at all. It sits between 95% and 100% from 2003 to 2024. The minutes never stopped printing the permit. Nor did the items disappear: the Commission heard 140 discretionary reviews in 2020 and 81 in 2022, well within its historical range even after the post-2013 decline in the DR share of the

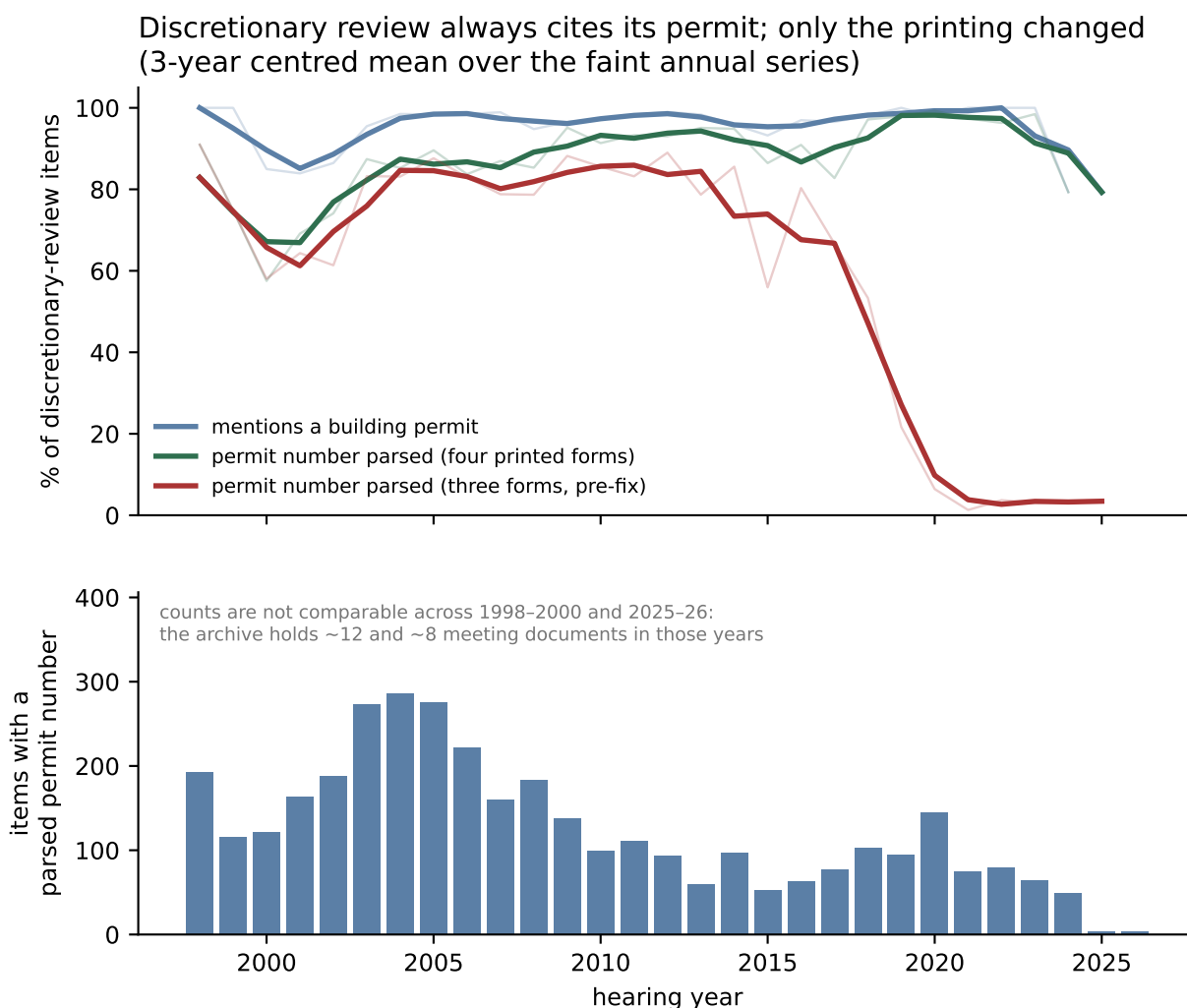


Figure 1: Discretionary-review items citing a permit, and the share from which a number can be parsed, with and without the fourth printed form. Rates are suppressed for 2025 and 2026, in which the archive holds 7 and 14 discretionary reviews respectively — the scrape reached the archive before the archive reached the present.

docket. What changed is the printing. From roughly 2017 the modern minutes render the permit as 2019.0618.3775 — year, MMDD, serial — with two separators where the older 2013.10.21.9832 has three. A regex written against the older form matches none of them, and the red line falls off a cliff it did not have to.

Adding the two-separator form (YYYY.MMDD.NNNN) recovers **711 items and 483 distinct permits**, at a 98.0% DBI match rate. Among discretionary reviews, parsed-permit coverage in 2020 goes from 9 items to 139, in 2021 from 1 to 75, in 2022 from 3 to 78. The permit route is usable for recent years, which is the years the paper most needs.

This is worth stating as a methodological point rather than a bug fix. The minutes are an archive assembled by a department that changed its word processor several times in twenty-eight years, and a deterministic rule that encodes one era’s formatting silently reports the other era as missing data. The check that caught it was cheap: measure the *mention* rate alongside the *parse* rate, and any gap between the two is a parser problem rather than a world problem. That pairing should be standard in this pipeline.

4 Which parts of the docket the permit route reaches

Table 3 cross-tabulates against `request_type`, and the answer is sharp enough to be worth being explicit about: **the permit route is a discretionary-review route**.

- **Discretionary review:** 4,117 items, 95.2% cite a permit, 85.5% yield a number, 83.8% match in DBI. This is expected — a discretionary review *is* a review of a filed building permit, so the permit number is the subject of the hearing — but the magnitude is the useful part.
- **Everything else:** at most 2% on any other family, and below 0.5% on conditional use, the largest single request type at 5,074 items. A conditional use is an authorisation to make a use of land; the building permit, if any, is filed afterwards and under a different number. The minutes have nothing to cite.

So the permit-number join buys a near-complete construction-outcome panel for one quarter of the docket, and nothing at all for the legislative, CEQA and historic families. Any analysis built on the permit join is conditioned on being a discretionary review, and inherits discretionary review’s own selection — small infill projects with a neighbour objecting.

5 What the match buys

Table 4 inventories what becomes observable. For a matched item:

- **The permit issued in 78.5% of cases**, a median of **245 days** after the hearing and 768 days at the 90th percentile. The hearing is not the end of the clock; it adds most of a year to it, and the right tail of that addition is as wide as the delay inside the hearing process itself.
- **Construction started for 38.5%**, dated by `first_construction_document_date`. That is the closest thing in the record to “something was actually built”.
- **32.9% were withdrawn, expired, cancelled, revoked or disapproved** and 48.9% completed. The abandonment states are the observable trace of a project that lost its bet — the outcome the corpus memo identifies as the only route to the selection problem, since projects never filed leave no record anywhere.

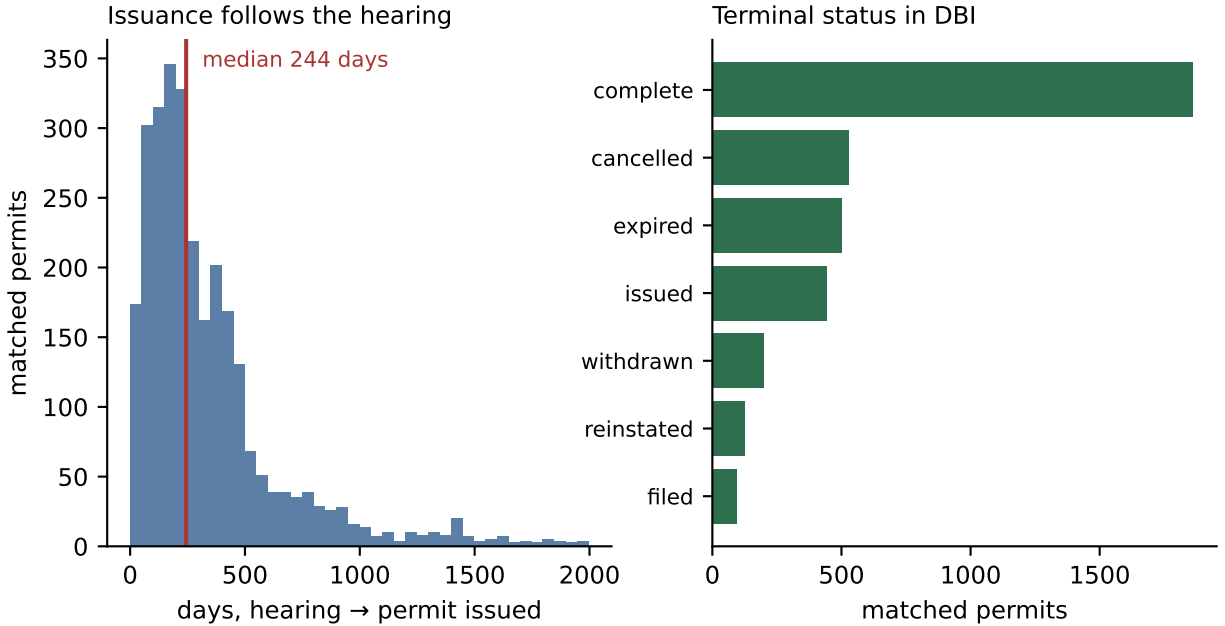


Figure 2: For the 3,801 matched item–permit pairs: elapsed days from the hearing to permit issuance (left, truncated at 2,000 days), and the permit’s terminal status in DBI (right).

- **Cost is recorded on 88.8%** (median revised cost \$350,000) and **proposed dwelling units on 84.8%**.
- **Supervisor district on 99.9%**, which is the cheapest available route to a political geography for the item table — the minutes print a district only in the modern era, and DBI prints one for essentially every permit.

The one comparison that does not work is units proposed at the hearing against units in DBI. The extraction schema has no unit-count field, and a regex over the request text recovers one for only 42 of the matched items, of which 57.1% agree with DBI exactly. That is too thin to be a validation of anything. It is not a defect of the join: discretionary reviews are predominantly additions and alterations to existing houses, and the minutes describe the work rather than counting units. If a unit count is wanted at the item level, DBI’s `proposed_units` on the matched subset is a better source than the minutes text.

6 The parcel route, scored honestly

For the three quarters of the docket with no printed permit number, the fallback is the parcel: 84.1% of items carry an assessor block and at least one lot, and DBI keys every permit the same way. The difficulty is that a parcel is not a project. The median matched parcel carries about ten permits, most of them over-the-counter alterations — 973,920 of DBI’s 1.3 million rows are `otc alterations permit` — and picking the wrong one attaches the wrong outcome to the hearing.

The permit-number subset is the ground truth for testing a rule, because on those items we know which permit the hearing was about. Table 5 scores seven rules on the 3,274 such items whose cited permit is present at the parcel DBI records for it. Coverage (does the rule return anything?),

precision (of what it returns) and recall (of what it should have found) are reported separately, because a rule that returns nothing has undefined precision and reporting it as zero flatters the wide rules.

The shape of the trade-off is clear. Taking every permit at the parcel has perfect recall and 0.14 precision — six wrong permits for each right one. Restricting to new construction, wood-frame new construction, alterations and demolitions — dropping over-the-counter work and signs — raises precision to 0.44 at almost no cost in recall, and is the single most valuable filter. Adding a date window trades recall for coverage: **filings from three years before to one year after the hearing return a candidate set for 94.0% of items, at 0.72 precision and 0.92 recall, with a median set size of one.** Tightening to one year before and six months after raises precision only to 0.75 while dropping coverage to 56% — the permit is often filed long before the hearing that reviews it, which is the whole reason a discretionary review exists.

Taking only the nearest filing gives point identification at 0.74 precision and 0.67 recall. That is the right rule if the analysis needs one permit per item and can tolerate a third of them being wrong; the three-year window, whose candidate set has a median size of one, is the right rule if the analysis can carry a set, for instance by taking the maximum project size or the earliest issuance across candidates.

Table 6 gives the reach. Of the 12,611 items with no printed permit number, 10,076 have a block and lot, 9,455 have a parcel DBI knows, and **6,664 — 41.1% of the whole docket — get at least one candidate permit** under the preferred rule. Combined with the 3,515 items matched on a printed number, some construction-side observation is available for **10,179 items, 62.8% of the docket**, of which a little over a third is exact and the rest is a small candidate set with a known error rate.

This is a positive result but it should not be oversold. The 0.72 precision is measured on discretionary reviews, because that is where the ground truth is, and discretionary reviews concern parcels with a specific pending permit. On a conditional-use item — a restaurant seeking a use authorisation on a busy commercial block — the parcel’s permit history is denser and less related to the hearing, and the rule will do worse. Extending the validation there requires ground truth that does not currently exist.

7 Conditions of approval are not in DBI

The corpus memo left open whether the conditions the Commission attaches at the hearing can be recovered from the permit record. They cannot.

DBI’s **description** field is populated on 98.6% of rows and describes the *work*: “install aluminum windows”, “infill right side lightwell. remodel (e) kitchen and add a new bathroom on 2/f of (e) building”. Searching all 1,295,048 rows for any mention of a condition of approval, a Planning Commission motion, or work performed pursuant to one returns **87 rows — 0.007%**. There is no field for planning conditions, and the free text does not carry them as a matter of practice. This is not a coverage gap that better matching would close; DBI records the permit’s own scope and status because that is what a building department regulates.

The consequence is that the conditions question has to be answered somewhere else, which is what the companion memo on conditions of approval takes up. The permit record answers “what happened to the project”, not “what was the project made to do”.

8 What this establishes, and what it does not

Established.

1. The permit-number join is a census, not a sample, and it holds at 97.6% across all four printed forms and all four decades.
2. The route is alive for 2017–2026; the earlier appearance of collapse was a parser artefact, now fixed and regression-visible in Figure 1.
3. The join is a discretionary-review instrument. It reaches 83.8% of DR items and under 2% of everything else.
4. For matched items, issuance, construction start, abandonment, cost, units and supervisor district are all observable, with issuance a median 245 days after the hearing.
5. A parcel-plus-window-plus-type rule extends partial coverage to a further 41% of the docket at 0.72 precision and 0.92 recall, validated against the permit-number subset.
6. Conditions of approval are absent from DBI.

Not established.

1. Whether the parcel rule’s precision survives outside discretionary review. It is measured where the ground truth is, and that is not a random part of the docket.
2. Whether an unmatched permit number means the permit does not exist or the minutes mis-transcribed it. 56 numbers, small enough to adjudicate by hand, and worth doing before any analysis treats non-match as an outcome.
3. Whether `proposed_units` in DBI is the right unit count for a project that was modified at the hearing. DBI records the permit as filed and as revised; which revision post-dates the Commission’s action is knowable from `status_date` but has not been checked.
4. Anything causal. The 245-day median lag from hearing to issuance is a description of a joined record, not an estimate of what the hearing did to the clock.

9 Reproducing this

`code/commission_minutes_processing/analyze_permits.py` does all of it: it caches DBI (`--refresh` to re-download), parses, matches, scores the parcel rules, writes both figures and `tables/permit_tables.tex`, and emits two artefacts next to the extraction on Dropbox — `permit_matches.csv`, one row per (item, cited permit) with the DBI fields attached, and `permit_summary.json`, which `analyze_corpus.py` reads so the corpus memo’s linkage table cannot drift from this one. No number in this memo is hand-placed.